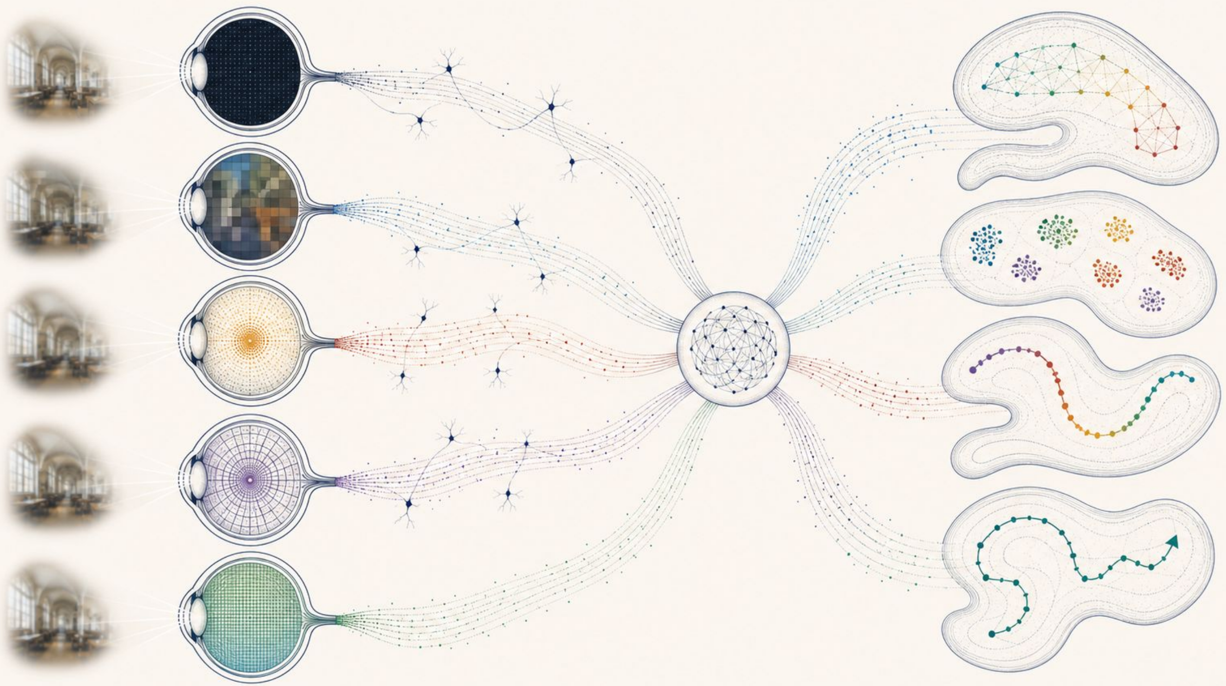
CS-503 VISUAL INTELLIGENCE
Final Presentation

Weilun Xu Rim Abkari
Léo Bruneau Zyad Aoutir

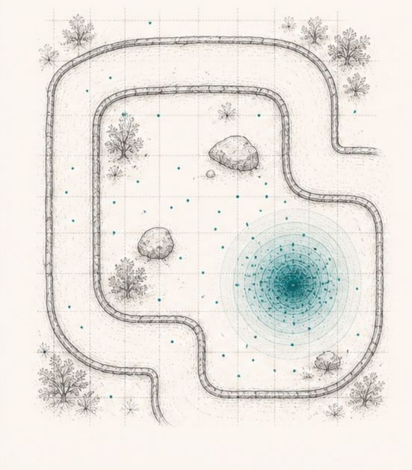
# Sensor Structure Shapes the Format of Cognitive Maps in Navigation Agents



## — MOTIVATION

# Spatial memory is not one format.

Biology suggests sensors shape the coordinate system.

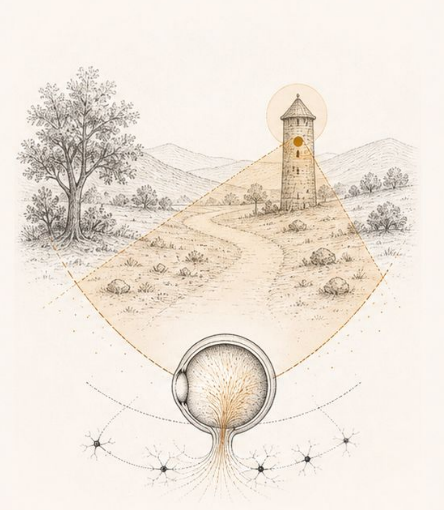


**world-centered**

**(place / grid-cell map)**

Where am I?

example: rats, bats

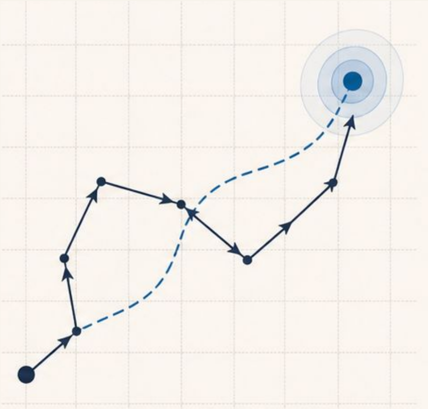


**view-anchored**

**(visual snapshot memory)**

What view locates me?

example: ants, bees



**motion-integrated**

**(path integration / home vector)**

How did I get here?

example: desert ants

**Our test: change only visual bandwidth and ask whether memory format shifts.**

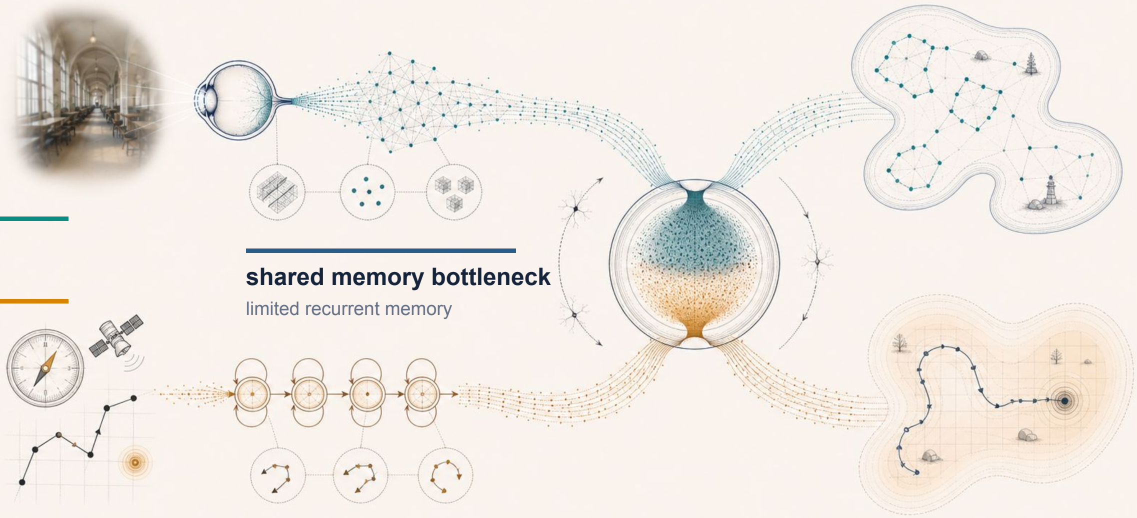
HYPOTHESIS

# Two routes compete for finite memory.

Position can come from per-step vision or from integrated sensor history.

**visual route**
per-step scene anchor

**integration route**
GPS + motion history

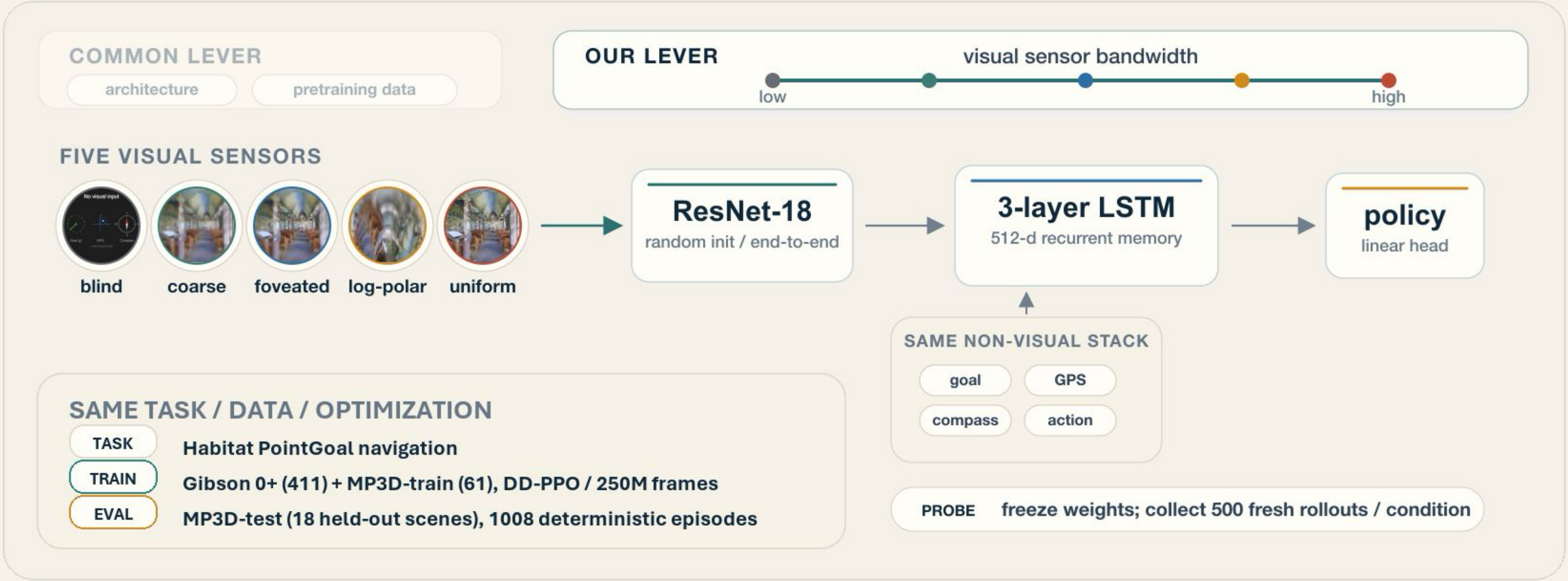


**shared memory bottleneck**
limited recurrent memory

**Capacity allocation hypothesis:** richer vision tilts memory from integration toward visual features.

# We change the sensor (and hence encoder), but keep agent the same

A matched PointGoal chassis isolates visual bandwidth as the experimental lever.



RESULT 1 / MAGNITUDE

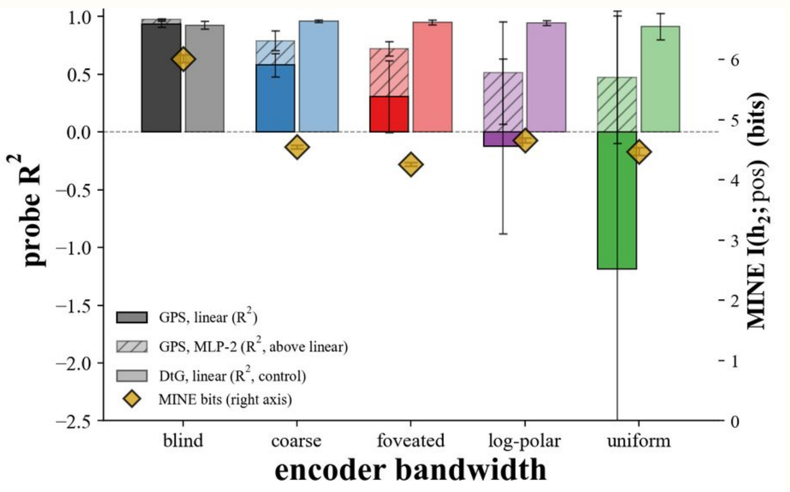
# Behavior converges; linear GPS memory collapses.

TABLE 1 / HELD-OUT NAVIGATION

## Behavior converges.

All five agents reach goals on held-out MP3D scenes.

| condition | success | SPL | steps |
|---|---|---|---|
| blind | 0.765 | 0.448 | 564 |
| coarse | 0.931 | 0.722 | 255 |
| foveated | 0.936 | 0.780 | 210 |
| log-polar | 0.942 | 0.767 | 239 |
| uniform | 0.937 | 0.789 | 232 |

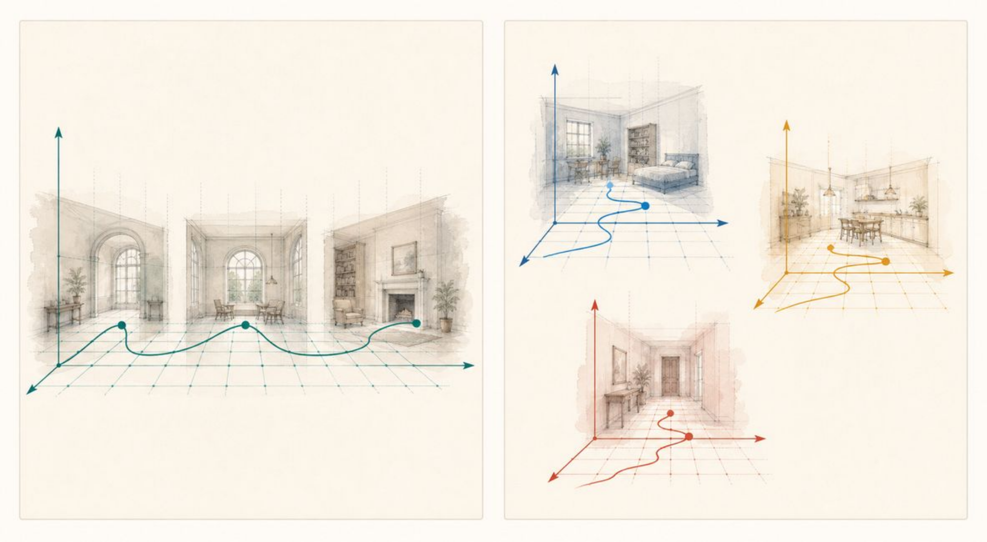



**richer vision preserves navigation while moving position out of a simple linear memory axis.**

RESULT 2 / FORMAT

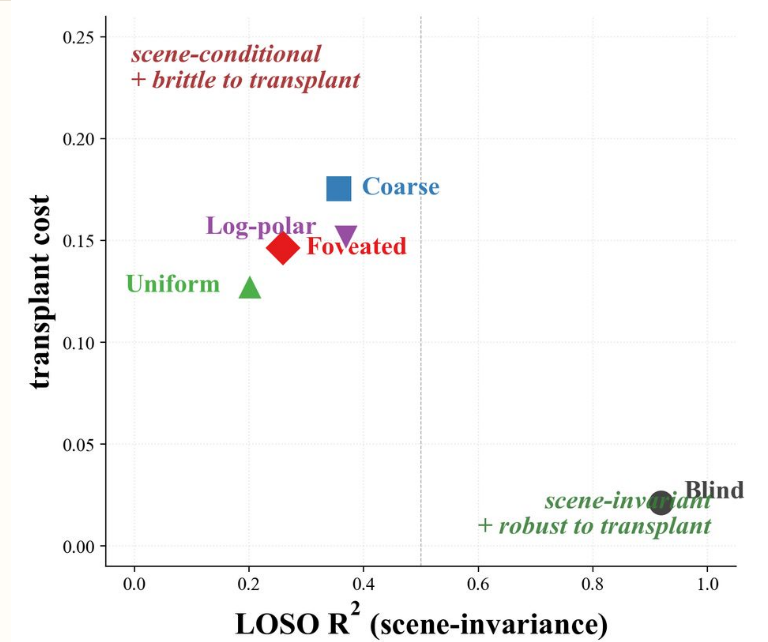
# Any vision makes memory scene-bound.

Position remains present, but the coordinate system stops transferring across rooms.



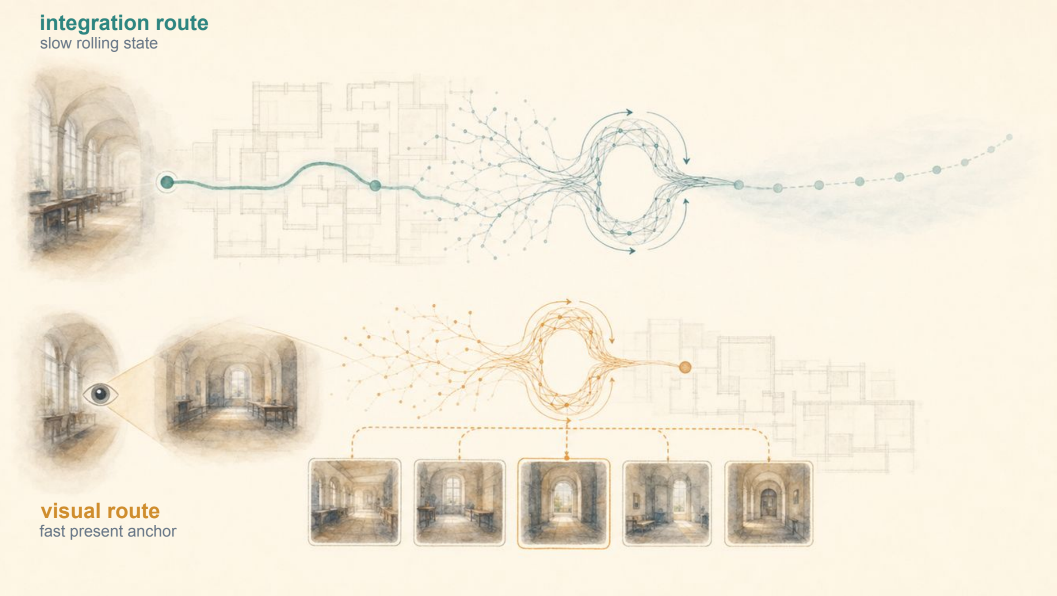
**blind**
one global coordinate system

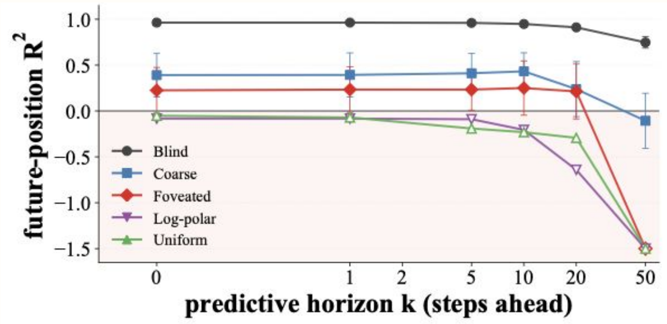
**rich vision**
separate scene-local systems



# Blind carries a forward-rolling trajectory.

Vision anchors the present; integration predicts positions many steps ahead.



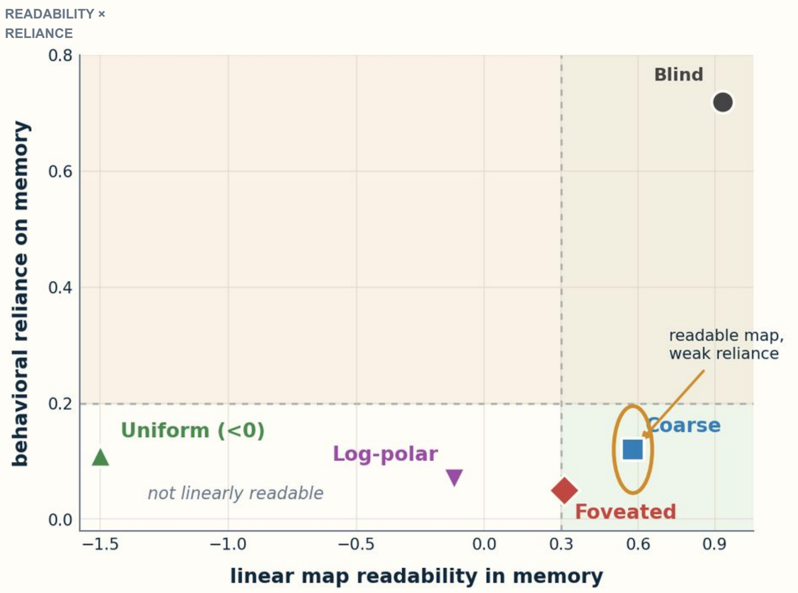

**Bottleneck Sensor (Blind, Coarse)**
readable at 50 steps

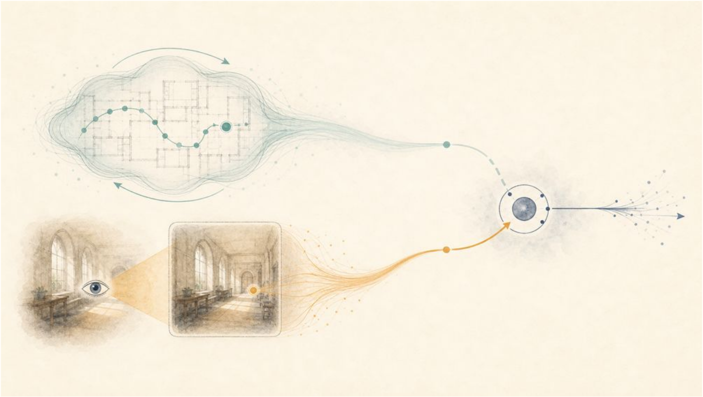
**Rich Sensor (Uniform, Log-Polar, Foveated)**
collapses toward chance

**Temporal shift: integration carries a future trajectory; vision refreshes the present.**

# Vision makes memory reliance optional.

A readable map can exist, but behavior may follow the visual route.



**Blind** no visual anchor; memory route drives behavior

**Coarse** map is readable, but vision can take over

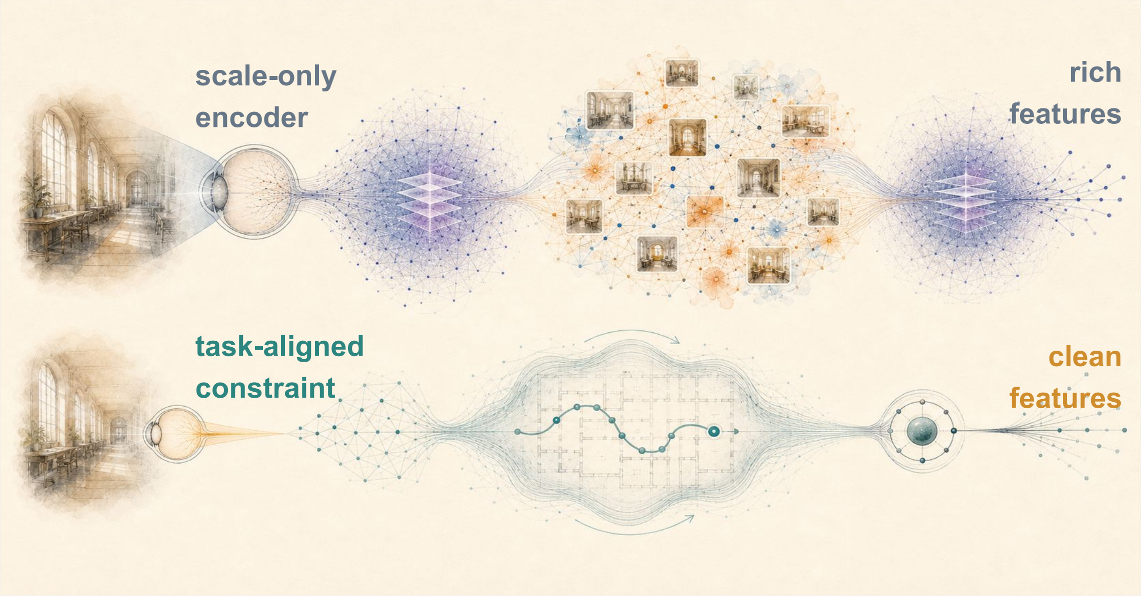
**Sensor structure changes both the map stored in memory and the route used for navigation**

## Broader Implication **Perception is part of the cognitive architecture.**

**Sensor/encoder structure changes the spatial route learned by memory and policy:**

when vision has enough information, explicit positional information become pass-through

## Future work



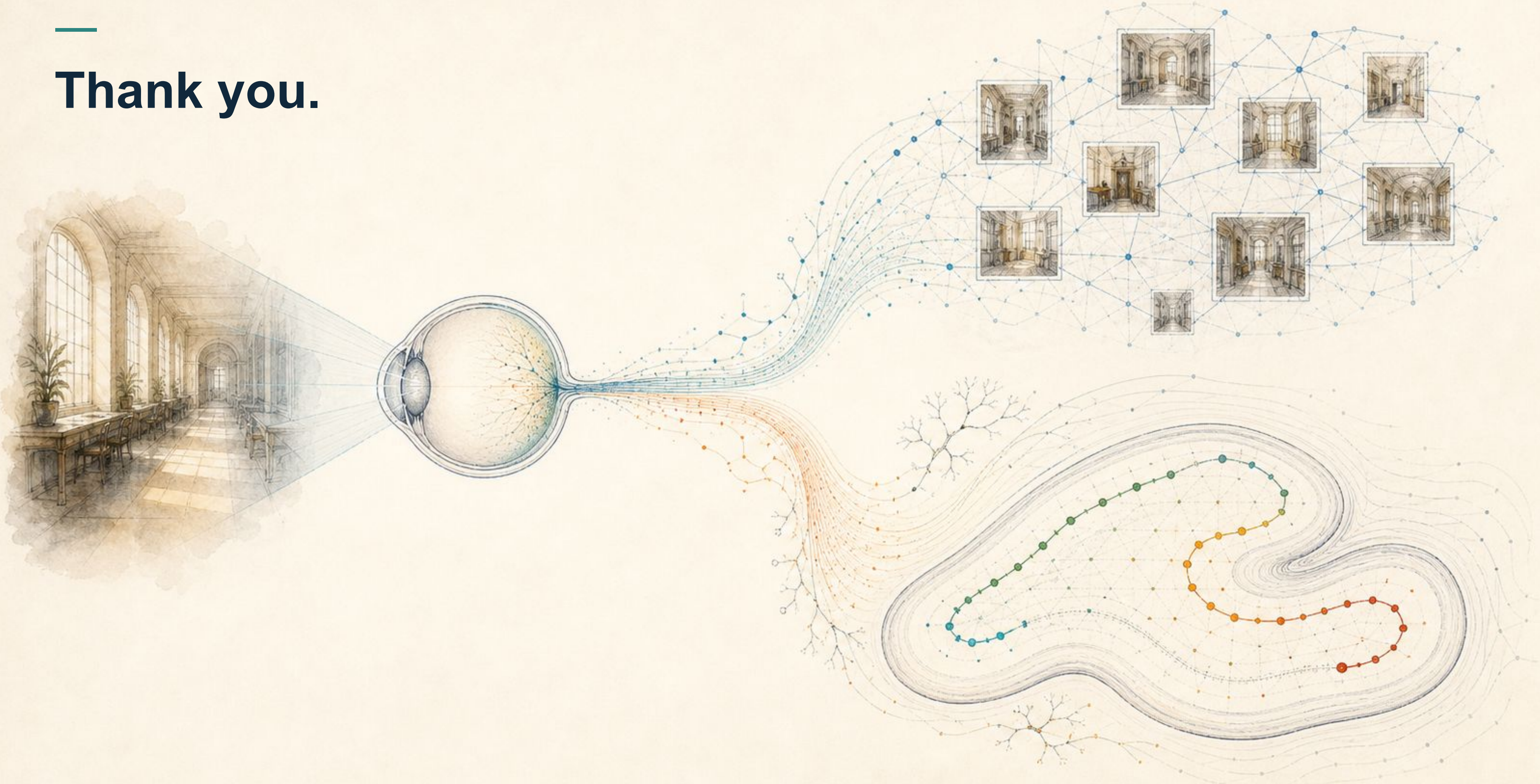

**(Broader Impact on visual intelligence)**

**The VLM Reasoning Bottleneck hypothesis:**

Rich visual encoded features are not aligned with the task.

**Task-Conditioned Encoding: task-dependent filtering of input visual features**

# Thank you.



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